

Privacy and Security in Distributed Data Markets

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Abstract

Data markets play a pivotal role in modern industries by facilitating the exchange of data for predictive modeling, targeted marketing, and research. However, as data becomes a valuable commodity, privacy and security concerns have grown, particularly regarding the personal information of individuals. This tutorial explores privacy and security issues when integrating different data sources in data market platforms. As motivation for the importance of enforcing privacy requirements, we discuss attacks on data markets focusing on membership inference and reconstruction attacks. We also discuss security vulnerabilities in decentralized data marketplaces, including adversarial manipulations by buyers or sellers. We provide an overview of privacy and security mechanisms designed to mitigate these risks. In order to enforce the least amount of trust for buyers and sellers, we focus on distributed protocols. Finally, we conclude with opportunities for future research on understanding and mitigating privacy and security concerns in distributed data markets.

Keywords

Privacy, Security, Data Markets, Distributed Protocols

1 Introduction

Data markets [3, 10, 12, 19, 20] are increasingly critical for modern economies, enabling data sharing that fuels artificial intelligence (AI) models, personalized services, and research. Economists pool data from varied government agencies to—more precisely—assess causal effects of policy interventions [35, 36]; scientists use data from private and government labs to enable drug discovery and chemoinformatics [15, 74]; organizations use pooled data for intrusion detection [29]. However, extracting value from data markets is stymied by the necessary trust that must be established with data users and data providers. Most of the trust issues stem from privacy and security concerns that result from sharing data via data markets. Trust and privacy are paramount concerns due to:

- (1) **Rising Privacy and Security Awareness:** Users are more aware and concerned about data misuse, data manipulation attacks, and privacy violations [11].
- (2) **Increasing Regulatory Pressure:** Laws like GDPR mandate strict privacy compliance, which affects data market operations [7].

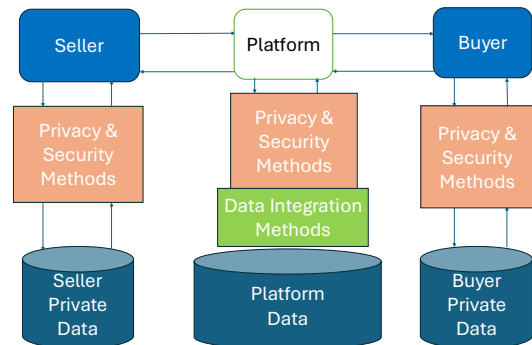


Figure 1: Overview of data flows in data markets that require privacy & security enforcements during data integration. For simplicity, we illustrate a single buyer and seller but there could be multiple sellers, buyers, and platforms in a data market.

- (3) **Need for Trust to Facilitate Data Sharing:** A lack of trust can lead to reluctance to share data, hindering the value that data markets can create [88].

In data markets, personal data is often anonymized or aggregated to protect privacy. However, with advanced machine learning techniques and increased computational power, re-identifying individuals from supposedly anonymized data is becoming easier. This tutorial covers the core privacy and security concerns in data markets, including the definitions and implications of privacy attacks and the mechanisms designed to mitigate these risks [14, 62]. Data markets are online platforms where data is exchanged, bought, or sold. They have become central to industries such as marketing, healthcare, finance, and technology, driving decisions with predictive models, customer insights, and strategic planning. However, the commodification of data raises important questions about privacy. Typically, a data market involves one or more buyers, sellers, and platforms. Every privacy and security mechanism assumes a **specific threat model** for an adversarial attack [68]. The platform is responsible for ensuring that the threat models in the data market are enforced and communicated to the buyers and sellers.

Figure (1) illustrates data flows between three entities: the **Seller**, the **Platform**, and the **Buyer**, with privacy and security considerations embedded at various stages to enable data flows:

- **Seller $\Leftrightarrow$ Platform:** The seller determines that they have valuable data to sell to a platform. Before the flow of data begins between the seller and platform, a series of interactions occurs to determine if the data is in the right format, is valuable enough, and meets the privacy, regulatory, fiduciary, or commercial requirements for the platform and seller. Then a contract is signed (e.g., via cryptographic commitment schemes [68]) that allows for data exchange.
- **Buyer $\Leftrightarrow$ Platform:** The buyer determines that they wish to buy valuable data through the platform. Before the flow of data begins between the buyer and platform, a series of interactions occurs to ensure that the privacy, regulatory, fiduciary, or commercial requirements for the parties are met. The buyer acquires the data after signing a contract and, if necessary, applies its own privacy and security controls before storing the data.
- **Within the Platform:** The platform performs data integration methods to combine, process, or analyze the data (e.g., in the form of tabular data) it has acquired from one or more sellers. This process may involve handling sensitive data, necessitating robust privacy and security measures. If the necessary privacy protections have already been applied to the data, the seller only has to perform data integration.
- **Within the Buyers and Sellers:** Depending on the requirements of each buyer or seller, additional privacy controls might be applied to the data before acquisition or transfer of data.

This tutorial addresses the urgent need for reliable frameworks to establish trust and privacy, making data markets safer and more attractive to stakeholders. Furthermore, we discuss privacy attacks and mechanisms to alleviate the possibility of such attacks.

1.1 Intended Audience

This tutorial is intended for:

- (1) **Researchers and Practitioners** in data management, machine learning, and cybersecurity who are interested in understanding the interplay between trust, privacy, and data monetization.
- (2) **Data Market Operators** who need to understand the privacy implications and trust requirements of their platforms.
- (3) **Policymakers and Legal Experts** looking to understand technical privacy solutions to inform regulation.

The tutorial assumes no prior background on privacy mitigation strategies, but assumes basic mathematical maturity.

1.2 Structure of Tutorial

The tutorial is structured as a 3-hour long tutorial and will cover the following core topics:

- (1) **Survey on Data Markets:** Introduction to data markets, their significance, and typical models of operation; overview of challenges in pricing and strategic behavior within data markets; current trust and privacy concerns in data exchanges. See Section 2.
- (2) **Privacy and Security Risks:** Discuss types of data privacy and security risks; analyze case studies of privacy breaches

in data markets, including breaches in healthcare, marketing, markets, and genomic data markets. See Section 3.

- (3) **Mechanisms to Minimize Risk:** Introduce audience to privacy-preserving technologies, such as k -anonymity, differential privacy, and secure multi-party computation; discuss solutions to mitigate security vulnerabilities within data markets. See Section 4.
- (4) **Regulatory Considerations:** Overview of privacy regulations affecting data markets; discuss compliance challenges and their impact on data market operations. contextualize GDPR and CCPA regulations. See Section 5.
- (5) **Open Problems & Future Work:** Exploration of ongoing challenges and research areas that are ripe for further exploration. See Section 6.
- (6) **Q&A and Open Discussion:** Interactive session for participants to discuss real-world applications and challenges.

2 Survey on Data Markets

Initial ideas on data marketplaces started in the early 2000s. Azure DataMarket is one of the first online marketplaces that was started in 2011. However, this market was a failed attempt and had to shut down in 2017 due to a lack of sustained customer interest. Since then, there have been many other platforms like Snowflake marketplace, Dawex, AWS, etc. These have been more successful but still struggle to attract a wide variety of users. We will start the tutorial with an overview of the landscape of these platforms and then go deeper into the specific challenges for distributed marketplaces. We will then dive into some main topics: (i) Centralized and Decentralized marketplace design (ii) Data discovery and acquisition.

2.1 Data Marketplace Design

Centralized Data Marketplaces [12, 49, 58, 64, 125] rely on a central authority to manage operations, including data storage, discovery, valuation, and transactions. They offer simplicity, efficient dispute resolution, and robust control over data quality. However, centralized systems face significant drawbacks, such as a single point of failure, limited transparency, and potential misuse of participant data. Decentralized data marketplaces (DDMs) are designed to facilitate the sharing, trading, and monetization of datasets or data-driven services in a trustless and distributed manner [71, 75, 112, 125]. By removing the need for a central authority, these marketplaces empower data providers (sellers) and data consumers (buyers) to interact directly, potentially increasing transparency, reducing costs, and enabling more equitable data exchanges. However, with this decentralized model come new challenges—particularly regarding how to ensure data security, protect user privacy, and maintain trust in the absence of traditional intermediaries. We will give an overview of both settings and discuss the following key components that comprise a marketplace.

2.1.1 Key Components of a Data Marketplace. Data marketplaces, centralized or decentralized, operate through a structured series of steps [125], including data acquisition (discovery, valuation, negotiation, and delivery), each facilitating seamless interactions and data transactions.

(1) **Data Registration:** This is the first step where sellers register their datasets along with metadata [21]. This registration helps the

central server to enable the discovery of datasets based on the user's query. Methods like blockchain-based commitments and privacy-preserving techniques (e.g., differential privacy) are often used to ensure authenticity and confidentiality [62, 64, 101].

(2) **Data Discovery:** Buyers often have an information need, such as training a machine learning model, performing a detailed analysis, or validating an experimental hypothesis. The data discovery component takes the user's requirements as input and returns a list of candidate datasets that fit their need [30, 48, 51, 58, 62, 80, 91]. The majority of these techniques can be categorized into three groups based on how users express their search needs, (i) keyword-based techniques [30, 48] (ii) example-based search [94, 121] (iii) task-based search [51, 63]. We will cover these categories in more detail, during the tutorial.

(3) **Data Valuation:** Once the relevant dataset has been identified, buyers have to pay an appropriate price to get access. Datasets are valued using various methods like fixed pricing, auctions, or AI-driven utility scores [12, 125]. Collaborative approaches, such as federated analysis are often used to enhance fairness and accuracy in valuation [45, 71, 89, 103, 117, 119].

(4) **Negotiation:** In many cases, the sellers often negotiate pricing and terms of use before providing access to the datasets [18, 66, 79, 92, 93, 119, 126]. Negotiation in centralized marketplaces [66, 92, 93] relies on platform-mediated tools for efficiency, while decentralized marketplaces [18, 119] use peer-to-peer methods like smart contracts, prioritizing transparency but adding complexity. Both aim to finalize terms like pricing and usage rights.

(5) **Data Sharing:** Data is delivered to buyers through various methods, including centralized storage systems, decentralized networks, real-time streaming, or federated learning models. Ensuring the integrity of delivered data is critical, often achieved through cryptographic proofs or similar verification mechanisms [83, 102]. Sellers use tools like digital rights management (DRM) and watermarking to protect their data [34]. Compliance is monitored through smart contracts and blockchain audits, which help detect and prevent violations, ensuring fair and secure transactions [102, 116].

(6) **Payment:** In centralized marketplaces, like Google's platforms, payments are managed through integrated systems, offering simplicity, multiple payment options, and built-in dispute resolution. In decentralized marketplaces, payments rely on cryptocurrencies, escrow mechanisms, or blockchain for transparency and trust [98].

(7) **Dispute Resolution:** Sharing of data through a marketplace requires a leap of faith from buyers as they do not get access to the dataset until the money is paid. Due to this reason, the reality may sometimes not match their expectations, leading to a dispute [33]. Centralized marketplaces [99] rely on a central authority to manage disputes through streamlined processes, standardized policies, and customer service teams. In contrast, decentralized marketplaces lack a central authority and depend on blockchain-based mechanisms, such as smart contracts, reputation systems, and community-driven arbitration.

The above-mentioned points provide an overview of the components that are present in both centralized and distributed marketplaces. However, several features of decentralized marketplaces are unique. For example, the metadata shared by a seller may not match the actual dataset. Furthermore, privacy vulnerabilities stem

from the leakage of indirect information, such as metadata or computation results, even when raw data remains protected. Security risks include malicious behaviors like data poisoning and valuation manipulation, which distort marketplace operations. The lack of a central authority complicates dispute resolution, making it difficult to verify claims or mediate conflicts effectively and detect malicious behavior, as both buyers and sellers can act dishonestly. Finally, efficiency and scalability are hindered by the computational overhead of privacy-preserving techniques like secure multi-party computation, impacting system usability.

While these challenges make current frameworks impractical, these are fundamental research questions that the data management community is poised to solve. Table 1 highlights features and their related vulnerabilities. We provide more details about some of the privacy and security challenges in Section 3.

2.2 Data Discovery and Acquisition

Data discovery is the process of identifying relevant datasets that would help the buyer to uncover insights, patterns, or valuable information [30, 48, 51, 70, 80, 91, 94, 121]. It serves as a foundational step in any data-driven task, enabling organizations and individuals to locate relevant datasets that meet their specific requirements. In a centralized setting, all datasets from various sellers (or providers) are indexed by the central server which has access to all of these datasets. The central server maintains the index and relevant metadata to support the search. In a decentralized setup, each provider shares metadata like dataset characteristics, schema information, sample data points, and some summary of the dataset with the central server. The central server acts as a router to match buyers with relevant data providers. This matching process is often approximate as these are based on data summaries and not the actual data points. Data discovery is an active area of research in the data management community but still has several open questions to be studied. Prior techniques can be categorized into the following categories based on how users express their requirements.

- (1) **Keyword-Based Search:** Users search for datasets using descriptive keywords or tags associated with dataset metadata. This approach is simple and widely used but may lack precision for complex queries [30, 48].
- (2) **Example-Based Search:** Users provide examples of data or partial datasets, and the system identifies similar datasets. This method offers better specificity for certain tasks but requires robust similarity measures [94, 121].
- (3) **Task-Based Search:** Users define their objectives or tasks (e.g., classification, prediction), and the system matches datasets based on their relevance to the stated goal. This approach bridges dataset discovery with its intended application but can be computationally intensive [51, 63].

Due to the lack of a central server that maintains all datasets, distributed frameworks for discovery struggle with malicious attacks from sellers and buyers. For example, sellers may manipulate the metadata to make their datasets more appealing.

3 Privacy and Security Risks

As data markets grow, concerns around privacy and ethical use of data have come to the forefront. Since many datasets contain

Table 1: Features and Associated Vulnerabilities in Decentralized Marketplaces

Feature	Description	Vulnerabilities
Decentralized Ownership	Data ownership remains with the seller, and there is no central authority.	- Sellers may misrepresent data quality or characteristics. - Lack of oversight allows malicious actors to operate.
Metadata-driven Discovery	Buyers rely on metadata or descriptions to discover datasets.	- Metadata could be incomplete, misleading, or manipulated. - Sensitive information may be inferred from metadata.
Confidential Transactions	Data exchanges are facilitated without revealing the full dataset upfront.	- Indirect information leakage through shared samples, summaries, or statistics. - Risk of poisoned or tampered samples.
Blockchain-Based Smart Contracts	Transactions are governed by automated, immutable smart contracts.	- Bugs in smart contracts can lead to exploits. - Difficulty in adapting to evolving privacy or security requirements.
Anonymity of Participants	Participants can interact anonymously to protect privacy.	- Increased difficulty in verifying the authenticity and trustworthiness of participants. - Potential for Sybil attacks.
Distributed Storage	Data is stored across decentralized nodes rather than a central server.	- Risk of data tampering or loss from compromised storage nodes. - Challenges in verifying the integrity of distributed data.
Peer-to-Peer Negotiation	Buyers and sellers negotiate terms directly without intermediaries.	- Sensitive negotiation details may be leaked. - Potential for unfair practices or coercion.
Tokenized Payments	Transactions often use blockchain tokens or cryptocurrencies.	- Token volatility can affect marketplace stability. - Risk of financial fraud or payment disputes.
Federated Data Contribution	Data remains decentralized, with aggregated insights or models shared.	- Risk of malicious contributions affecting aggregated results. - Difficulty in validating the origin of data contributions.
Dynamic Data Availability	Real-time data or updates can be shared as it is generated.	- Increased risk of incomplete or inaccurate data being shared. - Time-sensitive vulnerabilities in real-time validation.
Open Participation	Anyone can participate as a buyer or seller in the marketplace.	- Increased likelihood of malicious participants. - Difficulty in regulating quality and trustworthiness.
Data Valuation Mechanisms	Data value is assessed based on its utility, relevance, or uniqueness.	- Sellers can manipulate valuation metrics (e.g., by crafting data to match buyer requirements). - Buyers may overpay for low-quality data.

personal or sensitive information, data markets must ensure compliance with privacy regulations, such as the California Consumer Privacy Act (CCPA) and related regulations which mandates that data brokers register with the state and disclose any data breaches [4, 9]. Some markets adopt differential privacy techniques, which limit the impact of any single data point on aggregate datasets, thereby enhancing privacy without compromising the dataset’s analytical utility [62]. Such methods are essential in maintaining consumer trust and safeguarding data subjects’ rights, especially as data markets expand into sensitive domains like health and finance.

3.1 Real-World Privacy Violations

We detail a few instances where privacy attacks were successful on data markets:

- (1) **Healthcare Data Markets:** Healthcare data is often shared between hospitals, researchers, and pharmaceutical companies for analysis. While such data exchanges are valuable for medical research, they come with significant privacy risks. For example, even anonymized medical records can be vulnerable to privacy attacks, potentially revealing individuals’ diagnoses or treatments. As a prominent example of a privacy breach in healthcare, consider the case where Sweeney [53, 104–106, 108] was able to join public voter registration lists with hospital discharge databases to identify William Weld (a governor of Massachusetts) in the dataset [109].
- (2) **Marketing and Consumer Data Markets:** Companies like Facebook and Amazon operate large-scale data markets where user data is collected and sold for targeted advertising [3, 6]. These companies aggregate massive amounts of personal information, including browsing history, purchase behavior, and demographic details. Privacy concerns arise from the potential for this data to be sold to third parties or combined with other datasets, leading to re-identification. A concrete example occurred with the Cambridge Analytica data scandal where the personal data of up to 87 million Facebook users were acquired without the users’ consent [100].
- (3) **Financial Data Markets:** Credit reporting agencies like Equifax collect and sell consumer financial data to lenders, employers, and others [5]. In 2017, Equifax experienced a major data breach, exposing the personal and financial information of over 140 million people [26]. This incident highlighted the risks associated with centralized data markets and the need for stronger privacy protections.
- (4) **Genomic Data Markets:** Platforms like 23andMe [1] collect genetic data from individuals for personalized reports and research. While genomic data has tremendous potential for advancing healthcare, it is also highly sensitive. If this data were to fall into the wrong hands, it could be used for discrimination (e.g., by employers or insurers) or other malicious purposes [37, 82, 95]. In 2008, Homer et al. [61] was able to use DNA data to perform statistical tests that

could identify if a user was in a released complex DNA mixture (containing allele frequencies) or not.

3.2 Privacy Attacks

There are a variety of attacks on data markets. In this tutorial, we will focus on membership inference and reconstruction attacks.

Membership Inference Attacks. Membership inference attacks are a type of privacy attack where the adversary attempts to determine whether a specific individual's data was part of the dataset used to train a machine learning model. This type of attack can have serious consequences, particularly when datasets contain sensitive information such as medical records, financial transactions, or personal preferences. For example, in a healthcare dataset, knowing whether an individual's data is part of a training set could expose sensitive information about their medical conditions [37, 82, 95]. Even if the dataset is anonymized, these attacks can still be performed by querying the model and observing its outputs. The underlying risk arises from the fact that machine learning models often behave differently for data points they have seen during training compared to data they have not.

Reconstruction Attacks. Reconstruction attacks occur when an adversary, with access to aggregate statistics or summary information, attempts to reverse-engineer individual data records. Even if only summary statistics like means, variances, or correlations are released, attackers can sometimes deduce the values of individual records [39]. Reconstruction attacks highlight the risks of publishing even seemingly benign aggregated data, as adversaries can exploit correlations in the data to infer sensitive attributes of individuals.

3.2.1 Privacy Vulnerabilities. Privacy attacks can lead to the following vulnerabilities within distributed data markets:

Metadata Leakage. This occurs when metadata exposes sensitive attributes, such as demographics or proprietary details [87, 124]. Adversaries can exploit rich metadata to infer dataset characteristics or link datasets from the same seller, potentially leading to privacy breaches, competitive disadvantages, and reduced trust in the marketplace's security.

Sample Exposure. It refers to privacy risks arising from the sharing of dataset samples or partial data to demonstrate utility [43, 54]. Adversaries may exploit these samples by employing reconstruction attacks, piecing together full datasets using statistical or machine learning techniques [22, 38, 50, 55, 59, 60, 78, 84, 122, 128]. Shared samples can also inadvertently reveal sensitive attributes or identifying details, posing risks to privacy. Additionally, buyers may misuse these samples without authorization, further compromising data privacy.

Query Privacy. It refers to the risk of breaching buyer privacy through the exposure of query details, interests, or negotiation information [127]. Sellers may infer proprietary strategies from these queries, while third parties could analyze observed patterns to gain insights into buyer activity.

Participant Identity Privacy. This involves protecting the anonymity and confidentiality of participants, but vulnerabilities such as de-anonymization attacks [25, 86] can link pseudonymous identities to real-world entities. Behavioral profiling [56] may expose identities through transaction patterns, and identity exposure enables trust exploitation, where adversaries target participants with tailored attacks. These issues can lead to targeted attacks, reputational damage, and reduced participation due to fears of identity exposure.

Transaction Privacy. It refers to protecting sensitive information related to the terms, value, or content of transactions. Vulnerabilities arise from blockchain transparency, where public ledgers expose transaction details such as prices and dataset hashes, and from eavesdropping on communications, revealing negotiation specifics [115]. Additionally, adversaries may analyze transaction patterns to estimate dataset value or utility. These issues can lead to competitive disadvantages for participants and the misuse of transaction insights to manipulate marketplace dynamics.

3.3 Security Vulnerabilities

Security vulnerabilities in decentralized data marketplaces (DDMs) arise across multiple dimensions, including the buyer side, platform side, and seller side [31, 57, 119, 120, 125]. On the buyer side, malicious data or adversarial manipulation can degrade model performance or mislead decision-making. On the platform side, vulnerabilities such as manipulation of rankings, valuation mechanisms, or query inference attacks can disrupt trust and fairness. On the seller side, dishonest behaviors from malicious sellers, such as providing tampered or low-quality datasets, undermine marketplace integrity and erode buyer trust, making it harder for honest sellers to compete and maintain credibility.

Data Quality and Integrity. These ensure that datasets in the marketplace are genuine, unaltered, and of high quality. Vulnerabilities include data poisoning [110], where sellers introduce malicious data to degrade model performance, misrepresentation of dataset quality or relevance, and overfitting traps [123], where datasets perform well under narrow evaluation criteria but fail to generalize. These issues can lead to buyers training unreliable models, wasting resources, and losing trust in the marketplace due to repeated dissatisfaction.

Marketplace Manipulation. It refers to the exploitation of marketplace mechanisms to gain unfair advantages, such as dominating dataset selection or distorting valuations [31, 40, 57, 119, 120]. Vulnerabilities include ranking exploits, where sellers manipulate metadata or features to influence ranking algorithms, Sybil attacks [40], where multiple fake identities flood the marketplace, and collusion among sellers to manipulate pricing, rankings, or buyer perceptions [119]. These actions overshadow legitimate sellers and lead to biased or suboptimal recommendations, eroding trust and fairness in the marketplace.

Transaction and Delivery Reliability. These guarantee that buyers receive the agreed-upon datasets and sellers are fairly compensated [47, 52]. Vulnerabilities include dataset non-delivery, where sellers fail to provide the promised data after payment, data substitution, where sellers deliver datasets different from those advertised, and

payment fraud, where buyers refuse to pay after receiving the dataset. These issues increase disputes between participants and necessitate robust mechanisms for effective dispute resolution within the marketplace.

Scalability and System Resilience. These ensure that the marketplace remains functional and efficient under high participation or targeted attacks. Vulnerabilities include denial of service (DoS) attacks [96], where malicious actors overwhelm the system with fake datasets or queries, performance bottlenecks caused by high demand or adversarial actions, and data overload, where irrelevant or low-quality datasets flood the marketplace. These issues reduce system responsiveness, frustrate participants, and make legitimate datasets harder to discover, diminishing the marketplace’s overall utility.

Dataset Free-Riding. Buyers may exploit data samples or gradients provided during valuation to train models without compensating sellers, bypassing the need to complete a purchase. These practices undermine sellers’ control, devalue their contributions, and erode trust in the marketplace, leading to financial and reputational losses while weakening the overall ecosystem.

Distributed Dispute Resolution. This exposes critical vulnerabilities that threaten the system’s robustness. Without a trusted mediator or access to ground truth data, sellers can exploit the system by delivering tampered or low-quality datasets, while buyers may falsely claim non-receipt or poor quality to avoid payment. Collusion between parties, such as fabricating disputes, further undermines trust and distorts marketplace operations. Additionally, the lack of scalable mechanisms makes the system susceptible to exploitation, as malicious actors can overwhelm dispute resolution processes, delaying outcomes and increasing costs.

4 Mitigating Risks

4.1 Security Risks

4.1.1 Definition of Security in Decentralized Data Marketplaces. A secure decentralized data marketplace (DDM) is a platform that ensures robust, reliable, and trustworthy interactions among participants, even in the presence of adversarial behaviors or unforeseen circumstances. Security in a DDM implies the following:

- (1) **Resilience to Adversarial Actions:** The system maintains functionality even under threats such as marketplace manipulation or data poisoning [17, 23, 27, 28, 46, 65, 67, 85, 111].
- (2) **Integrity of Data and Processes:** Data quality, valuation mechanisms, integrity checks (e.g., via watermarking [13]) and transaction workflows are robust against tampering and manipulation.
- (3) **Fairness and Protection:** The marketplace guarantees equitable participation and safeguards against exploitation by malicious actors.
- (4) **Transparency and Accountability:** All actions, including transactions and data updates, are traceable through verifiable logs (e.g., blockchain), ensuring participants are held accountable for their actions.

Security is critical to maintaining trust in DDMs, as participants often interact without prior trust relationships, relying on decentralized mechanisms to ensure security and fairness.

4.1.2 Mitigation Methods.

□ *Data Validation and Filtering.* A first line of defense is to validate data against poisoning and backdoor attacks before it is fully integrated into the model or released to buyers. Following are some ways to achieve this:

- **Resilience to Adversarial Actions:** The system maintains functionality even under threats such as marketplace manipulation or data poisoning [17, 23, 27, 28, 46, 65, 67, 85, 111].
- **Outlier and Consistency Checks:** Examine distributions, correlations, and known “sanity checks” to flag anomalous tuples.
- **Automated Quality Metrics:** Require sellers to provide metadata or cryptographic commitments about the dataset’s statistical properties. Any significant deviation during on-chain or local validation can signal manipulation.
- **Incremental Acceptance:** Let buyers sample or validate small subsets of data first. Malicious sellers inserting poisoned examples may reveal themselves early if anomalies are detected.

□ *Robust Model-Training Protocols.* When multiple parties share training data or coordinate in a federated learning scenario, adopt robust training algorithms that resist outlier or adversarial data points. The following techniques could be used to achieve this:

- **Byzantine-Robust Aggregation:** Incorporate techniques (e.g., coordinate-wise median or trimmed mean) instead of naive averaging to minimize the effect of outliers or adversarial updates.
- **Poisoning-Resistant Optimization:** Use specialized optimizers that reduce vulnerability to manipulated gradients.

□ *End-to-End Cryptographic Verification for Data Consistency.* This unified cryptographic verification pipeline ensures data consistency throughout the entire marketplace transaction, preventing sellers from providing misleading discovery results and buyers from falsely disputing dataset correctness. Below are the steps of the pipeline:

- ① **On-chain Commitment:** Before any data discovery or transaction takes place, the seller commits publicly to a cryptographic representation (Merkle root or hash) of their dataset on-chain, binding them securely to this exact dataset.
- ② **Provable Discovery Results:** During data discovery, sellers return only samples, metadata, or statistical summaries accompanied by cryptographic proofs (such as Merkle proofs), confirming their direct derivation from the committed dataset. This guarantees alignment between discovery results and the delivered dataset.
- ③ **Signed and Encrypted Delivery:** Upon transaction, each delivered dataset chunk is signed by the seller, ensuring provenance and enabling immediate buyer verification against the original on-chain commitments.
- ④ **Buyer’s Verification and Acknowledgement:** Buyers independently verify each received chunk against the on-chain commitment. After verification, buyers generate cryptographic receipts

acknowledging successful validation, preventing subsequent false disputes.

⑤ **Zero-Knowledge Proof-based Arbitration:** Should any dispute arise, buyers or sellers submit cryptographic zero-knowledge proofs demonstrating inconsistencies between delivered data and on-chain commitments without exposing sensitive raw data. The arbiter resolves these disputes solely through cryptographic proofs, effectively safeguarding data privacy and ensuring transaction integrity.

4.2 Privacy Risks

4.2.1 Definition of Privacy in Decentralized Data Marketplaces. Privacy in decentralized data marketplaces (DDMs) refers to the protection of sensitive information related to participants (buyers and sellers) and datasets throughout all interactions in the system [125, 127]. This encompasses safeguarding personal and proprietary information, preserving the confidentiality of transactions, and preventing unauthorized access or misuse of data. By ensuring robust privacy protections, DDMs foster trust among participants, encourage broader adoption, and align with regulatory requirements like GDPR [90] and CCPA [69]. Privacy in this context guarantees the following:

- (1) **Dataset Privacy:** Sensitive attributes of datasets are not disclosed beyond what is agreed upon.
- (2) **Participant Privacy:** Personal or proprietary information about buyers or sellers remains confidential.
- (3) **Transaction Privacy:** Details of data transactions, such as queries or negotiation terms, are not exposed to unauthorized entities.

These principles collectively establish a foundation for privacy-preserving interactions in decentralized environments. To mitigate privacy attacks, various privacy-preserving techniques have been developed. Two of the most prominent are k -anonymity and differential privacy.

4.3 k -Anonymity

k -Anonymity [97, 107] is a property that ensures that each record in a dataset is indistinguishable from at least $k - 1$ other records with respect to certain identifying attributes. This means that for any given individual, their data cannot be singled out unless there are at least k individuals in the dataset who share the same set of attributes. To achieve k -anonymity, datasets are often generalized or suppressed. Generalization replaces specific data values with more general ones, such as replacing an exact birthdate with a birth year. Suppression involves removing certain identifying attributes altogether. While k -anonymity offers some degree of privacy, it is vulnerable to attacks like homogeneity attacks and background knowledge attacks [16], where an adversary with additional information might still be able to identify individuals within a k -anonymous dataset. For that reason, within data markets, k -anonymity has not been adopted as much as differential privacy.

4.4 Differential Privacy

Differential privacy [41, 42] is a more rigorous privacy definition. A mechanism is said to be differentially private if the inclusion or exclusion of a single individual's data in a dataset does not

significantly affect the output of a computation on that dataset. This ensures that an adversary cannot confidently infer whether the data of an individual was used in a given analysis, regardless of their prior knowledge. Some data markets have adopted differential privacy [72, 73].

It is now well-understood that modern cryptography tools (such as functional encryption and secure function evaluation) do not protect against reconstruction and membership inference attacks [114]. Informally, such tools ensure that the adversary does not learn anything other than the outputs of the functions that are computed. However, the function outputs could leak too much information (e.g., see Dinur-Nissim attack [39]). *Differential Privacy* provides a suite of definitions to ensure that database queries do not leak information to adversaries.

There are, by now, many definitions of DP including pure, approximate, Rényi, and zCDP. We provide a subset of these definitions below. It is sometimes more mathematically convenient to assume that the size of the dataset n is public. Other DP definitions (e.g., add-remove as opposed to change-one) can be used when we cannot assume that the size is public information.

For the definitions below, we say that two databases $\mathbf{x}$ and $\mathbf{x}'$ are neighboring, expressed notationally as $d(\mathbf{x}, \mathbf{x}') = 1$ for any $\mathbf{x}, \mathbf{x}' \in \mathcal{X}^n$, if $\mathbf{x}$ differs from $\mathbf{x}'$ in exactly one row.

Definition 4.1 (Differential Privacy [41, 42]). Let $\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{R}$ be a (randomized) mechanism. For any $\epsilon \geq 0, \delta \in [0, 1]$, we say $\mathcal{M}$ is (ϵ, δ) -**differentially private** if for all neighboring databases $\mathbf{x}, \mathbf{x}' \in \mathcal{X}^n$, $d(\mathbf{x}, \mathbf{x}') = 1$ and every $S \subseteq \mathcal{R}$,

$$\mathbb{P}[\mathcal{M}(\mathbf{x}) \in S] \leq e^\epsilon \cdot \mathbb{P}[\mathcal{M}(\mathbf{x}') \in S] + \delta.$$

If $\delta = 0$, then we say $\mathcal{M}$ is ϵ -DP, sometimes referred to as **pure** differential privacy. Typically, ϵ is a small constant (e.g., $\epsilon \in [0.1, 1]$) and $\delta \leq 1/\text{poly}(n)$ is cryptographically small.

Now we go into multi-party differential privacy.

4.5 The Definition for Multi-Party Computation

We now consider an extension of differential privacy to a multiparty setting, where the data is divided among some m parties $P_1, \dots, P_m$. For simplicity, we will assume that m divides n and each party P_k has exactly n/m rows of the dataset, which we will denote by $x_k = (x_{k,1}, x_{k,2}, \dots, x_{k,n/m})$. (Note the change in notation; now x_k is a sub-dataset, not an individual row.) We consider the case that P_k wants to ensure the privacy of the rows in x_k against an adversary who may control the other parties.

As in the studies of secure multiparty computation, there are many variants of the adversary model that we can consider:

- **Passive vs. active:** for simplicity, we will restrict to passive adversaries — ones that follow the specified protocol but try to extract information from the communication seen (also known as “honest-but-curious” adversaries). However, most of the upper bounds (on sample complexity, for example) in the existing literature are known to hold only for active adversaries.
- **Threshold adversaries:** we can restrict the adversary to control at most t parties for some $t \leq m - 1$. For simplicity, we will only consider the case $t = m - 1$. Consequently, we

may assume without loss of generality that all communication occurs on a broadcast channel, as the adversary would anyhow see all communication on point-to-point channels.

- **Computationally bounded vs. unbounded:** as in the basic definition of differential privacy, we will (implicitly) consider computationally unbounded adversaries. There are works that discuss the goal of satisfying differential privacy for computationally bounded adversaries [81].

For a more thorough study of multi-party differential privacy, see the survey by Vadhan [114]. A multi-party protocol proceeds in a sequence of rounds until all honest parties terminate. Informally, in each round, each party P_k selects a message to be broadcast based on its input x_k , internal coin tosses, and all messages received in previous rounds. The output of the protocol is specified by a deterministic function of the transcript of messages exchanged. As in secure multiparty computation, one can also consider individual outputs computed by the parties P_k , which may depend on their private input and coin tosses, but we do not do that for simplicity. Given a particular adversary strategy A , we write $\text{View}_A(A \leftrightarrow (P_1, \dots, P_m))(x)$ for the random variable that includes everything that A sees when participating in the protocol $(P_1, \dots, P_m)$ on input x . In the case we consider, where A is a passive adversary controlling $P_{-k} = (P_1, P_2, \dots, P_{k-1}, P_{k+1}, \dots, P_m)$, $\text{View}_A(A \leftrightarrow (P_1, \dots, P_m)(x))$ is determined by the inputs and coin tosses of all parties other than P_k as well as the messages sent by P_k .

Definition 4.2 (Multiparty Differential Privacy [24]). For a protocol $P = (P_1, \dots, P_m)$ taking as input datasets $(x_1, \dots, x_m) \in (X^{n/m})^m$, we say that P is (ϵ, δ) -**differentially private** (for passive adversaries) if for every $k \in [m]$ and every two datasets $x, x' \in (X^{n/m})^m$ that differ on one row of P_k 's input (and are equal otherwise), the following holds for every set T :

$$\Pr[\text{View}_{P_{-k}}(P_{-k} \leftrightarrow (P_1, \dots, P_m)(x)) \in T] \leq e^\epsilon \cdot \Pr[\text{View}_{P_{-k}}(P_{-k} \leftrightarrow (P_1, \dots, P_m)(x')) \in T] + \delta.$$

4.6 Two-Party Differential Privacy

Now let us look at the case of $m = 2$ parties, each holding $n/2$ rows of the dataset, which seems closer to the trusted curator case than to the local model. Indeed, in this model, any counting query can be computed with error $O\left(\frac{1}{\epsilon n}\right)$: each party just adds $\text{Lap}(1/(\epsilon \cdot (n/2)))$ noise to the counting query on her own dataset and announces the result; we average the two results to estimate the overall counting query. However, there are other simple queries where again there is a quadratic gap between the single curator ($m = 1$) and the 2-party case, namely the (normalized) inner product function $\text{IP} : \{0, 1\}^{n/2} \times \{0, 1\}^{n/2} \rightarrow [0, 1]$ given by $\text{IP}(x, y) = \langle x, y \rangle / (n/2)$. IP has global sensitivity $2/n$, and hence can be computed by a single trusted curator with error $O(1/n)$. But for two parties (one given x and one given y), the best possible error is again $\Theta\left(\frac{1}{\sqrt{n}}\right)$:

THEOREM 4.3 (2-PARTY DP PROTOCOLS FOR INNER PRODUCT [81]).

(1) *There is a two-party differentially private protocol that estimates IP to within error $O\left(\frac{1}{\epsilon\sqrt{n}}\right)$ with high probability.* (2) *Every two-party*

$(1, 0)$ -differentially private protocol for IP incurs error $\Omega\left(\frac{1}{\sqrt{n}}\right)$ with high probability on some dataset.

Constructing useful differentially private multiparty protocols for $m \geq 2$ parties is harder than constructing them in the standard centralized model (corresponding to $m = 1$), as a trusted curator could just simulate the entire protocol and provide only the output. An extreme case of multi-party differential privacy is when $m = n$, in which case the individual data subjects need not trust anyone else, because they can just play the role of a party in the protocol. This is called the local model. While this is the hardest model of distributed differential privacy, there are nontrivial protocols (namely, randomized response [32]).

Note that we have a better understanding of the local model (i.e., $m = n$) than for a smaller number of parties, such as $m = 2$. However, it still lags quite far behind our understanding of the single-curator model when we want to answer a set of queries (as opposed to a single query).

5 Regulatory Considerations

The regulatory landscape of data markets is increasingly shaped by data protection laws, which set standards for how personal data is collected, processed, stored, and exchanged. Key regulations like the General Data Protection Regulation (GDPR) [7] in Europe and the California Consumer Privacy Act (CCPA) [4, 9] in the United States are transforming data markets by imposing requirements for transparency, consent, and accountability in data transactions. These regulations prioritize the privacy rights of individuals, impacting how data markets operate and interact with both data providers and users.

GDPR: Setting a Global Privacy Standard. The GDPR, enacted in the European Union in 2018, is one of the most comprehensive data protection frameworks globally [7]. It applies to any organization handling the personal data of EU citizens, regardless of the organization's location.

GDPR's high penalties for non-compliance—up to 4% of an organization's annual global turnover—have driven many organizations globally to adapt to GDPR standards, impacting data markets both within and outside the EU. Data markets often implement GDPR-compliant practices across their operations to avoid penalties and build trust with EU-based clients.

CCPA: Strengthening Data Privacy in the U.S. The CCPA, introduced in California in 2020, is the U.S.'s strongest data protection law to date and has become a model for other states [4, 9]. While not as broad as GDPR, CCPA provides California residents with rights that reshape how data markets operate in the U.S. CCPA also includes provisions for statutory damages in cases of data breaches, incentivizing data markets to invest in security measures to avoid financial losses and protect consumer data.

Regulatory Impacts on Data Market Business Models. As privacy laws evolve, data markets must consider how regulatory compliance affects their business models. Many markets now prioritize **data aggregation and anonymized datasets** over personal data to mitigate compliance risks. Some data markets are shifting to focus on business-to-business (B2B) data transactions,

which typically involve aggregated, non-personal data, thus simplifying compliance requirements [2, 8]. Moreover, some markets have started exploring data escrow services as intermediaries to ensure regulatory compliance [118]. Data escrows act as neutral parties that store data securely and only release it when conditions meet regulatory standards, enhancing trust between buyers and sellers while safeguarding compliance. Ultimately, these regulations emphasize that data markets are responsible for protecting individual privacy, securing data, and fostering transparency. By prioritizing compliance with these standards, data markets can not only avoid legal pitfalls but also position themselves as trusted entities in an increasingly privacy-conscious world.

6 Open Problems & Future Work

As data markets continue to grow, so do the risks to individual privacy. Privacy attacks like membership inference and reconstruction demonstrate the need for privacy protection. To address these risks, techniques like differential privacy and multi-party computation have been developed, but each has its own limitations. As we move forward, it is crucial to balance the benefits of data sharing with the need for robust privacy protections.

Based on the analysis of vulnerabilities and challenges in decentralized data marketplaces (DDMs), we identify two overarching categories of research opportunities: Decentralized Trust Frameworks for Data Management, and Secure and Robust Data Discovery. These categories reflect the dual focus required to address the unique challenges posed by decentralization, including the need for reliable data handling, privacy preservation, and efficient discovery.

6.1 Decentralized Trust Frameworks for Data Management

Decentralized trust frameworks [44, 76] are critical for enabling secure and reliable interactions in systems without centralized authorities. Ensuring fairness, accountability, and confidence among participants, and fostering collaboration in trustless environments raise several open research questions. We list a few example questions below.

Data Consistency Management. Ensuring data consistency is fundamental to maintaining trust in DDMs, as it guarantees that the datasets buyers discover, evaluate, and use are identical to those registered and agreed upon [77]. In a distributed, trustless environment, challenges include malicious or unintentional alterations during transmission and integrity compromises during sampling or valuation. Verifying consistency across nodes without centralized oversight adds complexity, especially when minimizing computational and storage costs, making this a critical area for research and innovation.

Dispute Resolution Mechanisms. Dispute resolution in DDMs is a critical challenge, as conflicts between buyers and sellers—such as disputes over data quality, delivery, or usage terms—must be resolved without a central authority. The absence of intermediaries makes it difficult to verify claims, mediate disputes, or ensure fairness, particularly while preserving participants’ privacy. These challenges are compounded in trustless environments, where adversarial behavior or dishonest claims can further undermine the integrity of the marketplace. Addressing these issues is essential to

fostering trust, encouraging participation, and ensuring the equitable functioning of DDMs.

Privacy Auditing Frameworks. Privacy auditing frameworks are essential for ensuring compliance with regulations like GDPR [90] and CCPA [69] in DDMs [113]. A key challenge is verifying adherence to privacy policies while safeguarding participant confidentiality and dataset privacy. Effective frameworks must validate compliance without exposing sensitive information, striking a balance between transparency and privacy.

6.2 Secure and Robust Data Discovery

Data discovery in decentralized data marketplaces (DDMs) involves identifying, evaluating, and acquiring datasets. Development of privacy-preserving discovery techniques, and mechanisms to assess dataset value, and negotiate terms [91] are essential to protect sensitive information, prevent manipulation, and foster trust within the marketplace. We discuss some of these challenges below.

Privacy-Preserving Discovery. Privacy-preserving discovery [80] is critical for secure and efficient data discovery in DDMs while protecting sensitive information. Challenges include preventing adversaries from inferring buyers’ task objectives through queries and avoiding the leakage of sensitive metadata or dataset details during the discovery process. These vulnerabilities highlight the need for innovative solutions that balance privacy, efficiency, and usability, making it a compelling area for research and development in decentralized environments.

Robust Data Valuation Mechanisms. Robust data valuation mechanisms [57, 120] are crucial for maintaining fairness and accuracy in DDMs. Vulnerabilities such as adversarial manipulation, including data poisoning, can distort valuations and compromise trust in the system. Ensuring fairness is particularly challenging in methods like Shapley value or utility-based contributions, where datasets of varying importance must be evaluated equitably. These issues highlight the need for scalable and efficient solutions to support reliable and unbiased data valuation in distributed settings.

Seller Robustness Against Exploitation. Protecting seller-provided information during discovery is critical in DDMs. Malicious buyers may exploit valuation outputs, such as gradients or intermediate results, to extract sensitive information [129] or train models without completing a purchase. Excessive queries or computational abuse can further strain sellers’ resources, causing inefficiencies and financial losses. These vulnerabilities undermine the fairness of the marketplace and pose significant challenges to maintaining seller trust and participation.

Sybil Attack Detection and Prevention. Sybil attacks [40], where malicious participants create multiple fake identities to manipulate rankings or valuations, pose a significant threat to fairness in DDMs. Detecting such attacks is particularly challenging in decentralized systems, as it requires identifying fake identities without violating participant privacy. Collusion between Sybil accounts further exacerbates the issue, allowing coordinated efforts to distort rankings, valuations, and marketplace dynamics, ultimately undermining the system’s integrity and trust.

7 Presenter Biographies

Daniel Alabi is an (incoming) assistant professor in the Electrical and Computer Engineering department of the Grainger College of Engineering at the University of Illinois at Urbana-Champaign. Alabi's current research areas include privacy, cryptography, databases, and information theory. Before coming back to academia, Alabi was a database kernel engineer at MongoDB, Inc.

Sainyam Galhotra is an assistant professor in Computer Science at Cornell University and a field member for Computer Science, Statistics and Data Science. Previously, he was a Computing Innovation Fellow pursuing postdoctoral research at the University of Chicago. He received his Ph.D. from the University of Massachusetts Amherst. His research has been published in top-tier Data Management (SIGMOD, VLDB, PODS, & ICDE), AI (NeurIPS, AAAI & AIES) and Software Engineering (FSE) conferences. He is a recipient of the Best Paper Award in FSE 2017 and Most Reproducible Paper Award in both SIGMOD 2017 and 2018, and Best Artifact Paper Honorable Mention Award in SIGMOD 2023. He was recognized as a Data Science rising star, a DAAD AInet Fellow, and as the first recipient of the Krithi Ramamritham Award at UMass for contribution to database research.

Shagufta Mehnaz is an assistant professor of the Computer Science and Engineering department at The Pennsylvania State University. Her research focuses on enhancing the privacy and security of machine learning techniques and models themselves and developing novel machine learning techniques to protect data security and privacy. She directs the PRiVacy, Security, and Machine Learning lab (PRISMLab) at Penn State. She obtained her Ph.D. in Computer Science from Purdue University in 2020. She is a recipient of the Best Paper Award in ACM CODASPY 2017. She is a recipient of the Bilsland Dissertation Fellowship at Purdue and the Faculty For The Future Fellowship. Her research has been published in top-tier Security/Privacy (IEEE S&P, USENIX Security, PETS, NDSS) and AI/ML/Data Management (AAAI, ICDE) conferences.

Zeyu Song is a Ph.D. candidate in Computer Science at The Pennsylvania State University. His research interests focus on the security and privacy of machine learning, as well as leveraging machine learning techniques to address security challenges.

Eugene Wu is an associate professor in the Computer Science department at Columbia University. He is broadly interested in technologies that help users play with their data. His goal is for users at all technical levels to effectively and quickly make sense of their information.

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